

HA NGUYEN

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EDUCATION

McMaster University

Honours Computer Science Co-op (B.A.Sc.), Minor in Economics

Hamilton, Ontario

Sept 2021 – May 2026

- GPA: 3.5/4

- Relevant Courses: Data Structures & Algorithms, Natural Language Processing, Applied Machine Learning, Databases, Applied Cryptography, Operating Systems, Computer Networks.

EXPERIENCE

Software Developer Intern – Frontend and Backend

May 2024 – Aug 2025

Evertz Microsystems Ltd.

Burlington, Canada

- Integrated **AWS services including Lambda, S3, and DynamoDB** to build scalable, event-driven backend workflows supporting Evertz's cloud platform, EIO, used by broadcast clients across 100+ countries.
- Built and enhanced production frontend features for Evertz's cloud platform, EIO, using **Angular and TypeScript**, translating product requirements and UI designs into responsive, user-friendly components.
- Implemented feature logic, API data handling, validation states, and edge-case behavior while collaborating with backend and QA teams to prepare releases.
- Used **Postman** to manually test request/response contracts, simulate edge cases, and validate API behavior before frontend integration.
- Utilized **mabl** in CI/CD pipelines to develop robust end-to-end test suites, boosting test coverage by 90%.

Software Developer Intern

May 2023 – Aug 2023

Harmonysoft

Hanoi, Vietnam

- Built a full-featured management system handling data for up to 200 students, teachers, classrooms, and equipment, reducing administrative overhead by 70%.
- Developed application logic and data flow across web and mobile platforms using ReactJS and React Native, with HTML, CSS, and JavaScript, used Figma to design mobile applications.

PROJECTS

AI-Powered Real Time Sign Language | *Python, PyTorch, Vue, MediaPipe, NLP, Hugging Face*

Sept 2025 – Apr 2026

- Built a **real-time** ASL translation pipeline recognizing continuous signing from live video and converting it into structured gloss sequences using a **Temporal Graph Convolutional Network** with **sliding-window inference**.
- Fine-tuned/ Experiment with a **BART-based seq2seq** model, evaluating output with **BLEU and BERTScore** metrics, achieving 87% semantic similarity with custom conversational ASL glosses.
- Developed a **browser extension** for Zoom, Google Meet, and Microsoft Teams providing real-time ASL-to-English translation and live captioning for online meetings.
- Implemented **RESTful APIs and WebSocket** communication between frontend and backend services, supporting **real-time streaming inference** and concurrent users.
- Used **LLM tools** (Claude, ChatGPT, Cursor) to brainstorm architecture, structure frontend/backend code, compare implementation options, and accelerate iteration while reviewing outputs for correctness and maintainability.

The Dill Banh My | *TypeScript, AWS, ReactJS, Material UI, HTML, CSS*

Jun 2025 – Present

- Built a serverless **React** e-commerce app for Vietnamese bánh mì and drinks, originally hosted on **AWS Amplify** and now migrating to a raw AWS architecture.
- Re-architected backend deployment using **IAM roles, API Gateway, and Lambda**, with protected REST endpoints for admin menu CRUD operations and CI/CD.

EXTRACURRICULAR & INTERESTS

Cursor Hackathon Toronto | Track Winner, **NailedIt.ai**

2026

- Built a voice AI system that automates nail salon booking calls using English-accent conversational AI.
- Tech Stack: React, FastAPI, ElevenLabs, Twilio, Google Calendar API.

MakersLounge Innovation Hackathon Toronto | Track Winner, **snoop.ai**

2026

- Built an AI-powered tool that monitors competitors, summarizes market updates, and turns business signals into actionable insights.
- Tech Stack: Next.js, React, FastAPI, SQLite, C++, Groq AI.

CERTIFICATES

Microsoft Certified: Power BI Data Analyst Associate (Score: 803/1000) (Jan 2024 – Jan 2025)

SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, HTML/CSS, C

Libraries: ReactJS, React Native, NumPy, PyTorch, scikit-learn

Frameworks: Angular, FastAPI, REST API

Tools: Git, Jira, mabl, VS Code, Eclipse, Figma, AWS, Power BI, Word, PowerPoint, Excel, Outlook

Concepts: Data Structures & Algorithms, Software Design, CI/CD, Agile, AI/ML, NLP